

MechaScreener: Large Language Model-Based Automated Screening for Systematic Reviews and Research

Connor Forbes¹, Matt Carter¹, and Justin Clark¹

¹Bond University, Institute for Evidence-Based Healthcare

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Abstract

The screening of thousands of articles is the primary bottleneck in the systematic review (SR) process. This paper presents MechaScreener, an automated screening tool that utilises a Large Language Model (LLM) to rank article relevance based on PICOT or general eligibility criteria. Using an LLM, we developed a scoring mechanism (1–5) to triage references. Our evaluation across 10 diverse reference libraries demonstrates that MechaScreener achieves a perfect mean recall of 1.00 (100%) while maintaining an overall mean specificity of 0.61. This effectively reduces the manual screening workload by over 60% without missing any relevant studies.

1 Introduction

Systematic reviews (SRs) are considered one of the best ways to answer a research question; however, they are time-consuming, resource heavy and can be out of date by the time they are published. They require multiple, complex tasks to produce [1], one of which is screening the results of a comprehensive search of literature. This often results in thousands of irrelevant references [2], which must be manually screened by at least two reviewers [3]. This can be a time-consuming task as screening speed varies based on the team working on the review, the criteria and the software used, but can range from 60-300 references per hour [4].

Automation tools aim to reduce this burden by estimating the relevance of references for a given review. Large Language Models (LLMs) appear to be a promising method to achieve this, however recall continues to be a limiting factor [5]. To effectively automate screening, recall (the number of relevant studies included) needs to be near 1.00 (meaning all relevant studies were correctly identified). To date no LLM has been able to achieve a recall of 1.00 consistently across reviews, while reducing the screening burden in a measurable way [6].

To address the need of reducing the burden of screening for reviews we developed the MechaScreener. The MechaScreener is a LLM screening tool available via the Evidence Review Accelerator (TERA) [7]. It is a Zero-shot screening tool, meaning it requires no training data. To use it a reviewer simply uploads the references they want to screen, and enters the eligibility criteria for their review. This study outlines the development of the MechaScreener and its screening results across 15 reviews.

2 Methods

2.1 System Architecture

MechaScreener was developed using a Node.js/TypeScript backend environment with a Vue 3 frontend. The system processes reference citations in batches, extracting the title and ab-

struct and pairing them with a user-provided screening criteria string (formatted as Population, Intervention, Comparator, Outcome, Type of study (PICOT) criteria [8] or general inclusion/exclusion rules).

2.2 Prompting and Scoring

The MechaScreener system utilises a structured prompt consisting of:

1. **System Prompt:** Defined instructions for the LLM to act as a screening assistant.
2. **Reference Data:** Title and abstract of the paper.
3. **Criteria:** The eligibility criteria for the specific review.

In the context of Large Language Models, **temperature** is a parameter that controls the randomness or creativity of the model’s output distribution, while **top_k** restricts the model to sample only from the k most probable next tokens. To ensure the screening process is completely deterministic and highly reproducible for research purposes, we utilised a model configuration with a **temperature** of 0 and **top_k** of 1, effectively forcing the model to always select the single most likely response.

The model is instructed to provide a score from 1 (low probability of inclusion) to 5 (high probability of inclusion).

2.3 Experimental Design

The evaluation of MechaScreener was conducted in two phases:

- **Development Dataset:** Five control reference libraries were used to calibrate the prompt and scoring threshold to maximise recall and specificity.
- **Evaluation Dataset:** Ten additional reference libraries were used for independent validation of the tool’s performance across different medical and social science domains (e.g., COVID-19, diabetes, bone health). This includes 5 SRs that were looking for Randomised Controlled Trials (RCTs) and 5 SRs that were looking at non-RCT studies.

All 15 reference libraries are available online in our GitHub repository. The 10 evaluation studies were sourced from previous Cochrane reviews. As title/abstract screening results are not available, we used the final full-text inclusion decisions to determine what the relevant studies were. To compile the set of reviews used in the evaluation we reran searches from 10 randomly selected Cochrane reviews. This was done by searching PubMed for a Cochrane review, then generating a random number using the Google random number generator. This number was between one and the total number of search results found, e.g., if 1000 results were found the random number was between one and 1000. The search result that corresponded to this number was checked to see if it met our inclusion criteria of 1) having searched three or more databases; 2) reported search strategies for all databases; 3) have an identifiable list of included studies; 4) have a minimum of 6 to a maximum of 50 included studies; and for five of the reviews including randomised controlled trials and for the other five including study types other than randomised controlled trials. Any references missing abstract data were excluded from the evaluation.

2.4 Metrics

The performance of MechaScreener is evaluated primarily using two metrics: recall and specificity.

Recall (or sensitivity) measures the proportion of actual relevant studies that were correctly identified by the tool. In the context of systematic reviews, maximising recall is paramount to ensure no critical literature is missed. It is defined as:

$$\text{Recall} = \frac{\text{True Positives (TP)}}{\text{True Positives (TP)} + \text{False Negatives (FN)}} \quad (1)$$

Specificity measures the proportion of irrelevant studies that were correctly excluded. A higher specificity indicates a greater reduction in the manual screening workload, as fewer irrelevant articles are passed to the human reviewers. It is defined as:

$$\text{Specificity} = \frac{\text{True Negatives (TN)}}{\text{True Negatives (TN)} + \text{False Positives (FP)}} \quad (2)$$

2.5 Development Results

Five control libraries were used for development to maximise recall (number of references correctly included) and specificity (number of references correctly excluded). These libraries were compiled based on previous reviews conducted at the Institute for Evidence-Based Healthcare (IEBH). The developmental results for these libraries are presented in Table 1, with recall and specificity metrics for each library in Table 2.

Table 1: Results Across 5 Development Libraries

Library Name	No. of Refs	Include Studies Correctly Included	Exclude Studies Correctly Excluded
Antibiotic Prescribing and Telehealth [9]	934	13/13 (100%)	612/921 (66%)
Long Covid [10]	1,216	16/16 (100%)	552/1,200 (46%)
Natural History Primary Care [11]	5,913	11/11 (100%)	3,169/5,902 (54%)
Non Drug Interventions [12]	5,342	35/35 (100%)	3,972/5,307 (75%)
Salt Substitution [13]	889	23/23 (100%)	733/866 (85%)

Table 2: Mean Statistics for MechaScreener Across 5 Development Libraries

Library Name	Recall	Specificity
Antibiotic Prescribing and Telehealth	1.0000	0.6645
Long Covid	1.0000	0.4600
Natural History Primary Care	1.0000	0.5369
Non Drug Interventions	1.0000	0.7484
Salt Substitution	1.0000	0.8464
Mean	1.0000	0.6513

2.6 Evaluation Dataset

Ten reference libraries were used for evaluation of performance of MechaScreener. These ten libraries include a diverse range of topics shown below (5 studies that were RCTs and 5 studies that were non-RCTs).

Table 3: Overview of References and Missing Abstracts in Evaluation Libraries

Library Name	Total References	Missing Abstracts
RCT Studies		
Balneotherapy for Chronic Venous [14]	338	11
Calcium Vitamin D for Bones [15]	5,756	255
Methylxanthine for Apnea [16]	2,781	334
Phosphodiesterase 5 Inhibitors [17]	2,672	338
Topical and Oral Steroids for Om [18]	4,292	583
Non-RCT Studies		
Covid Schools [19]	2,162	21
Diabetes Tb [20]	5,177	165
Falls Prevention [21]	18,098	1,095
Fluoride Fluorosis [22]	11,829	2,527
Sanitation Diarrhoea [23]	10,770	297

3 Results

MechaScreener was tested across 10 evaluation libraries containing a total of over 58,000 references. Figure 1 summarises the performance metrics visually across each library and highlights the overall average for the evaluation set.

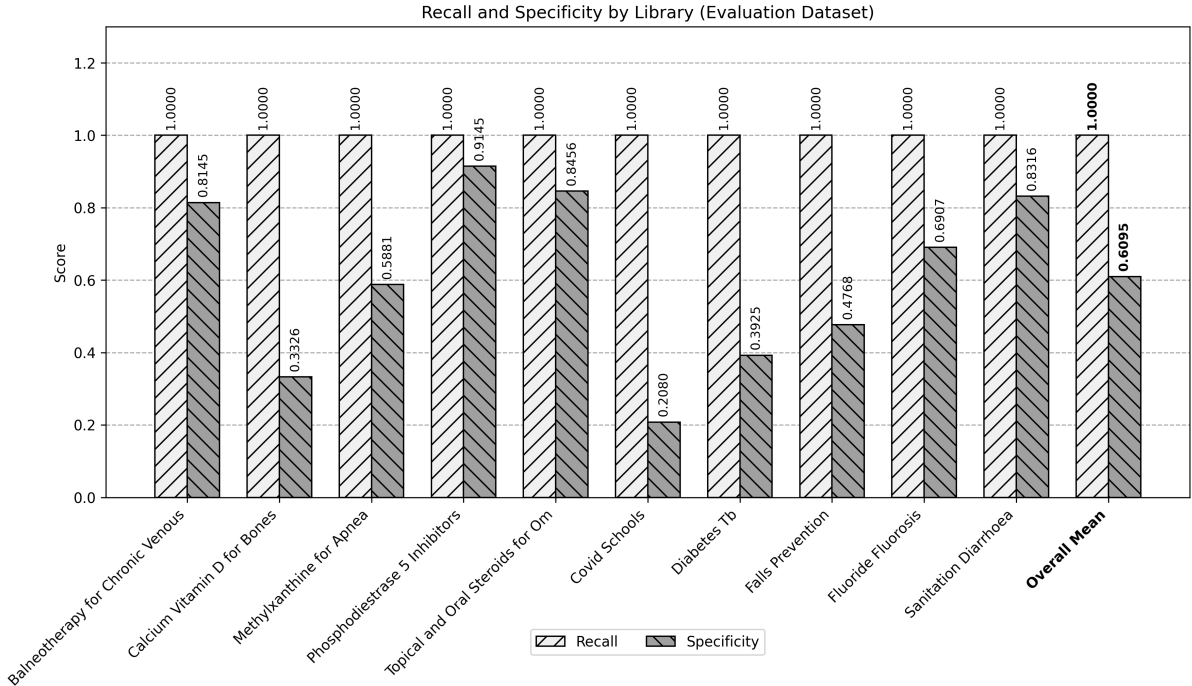


Figure 1: Recall and Specificity Across 10 Evaluation Libraries, Including the Overall Mean.

Within the evaluation dataset, the system achieved a perfect mean recall score (1.0000), meaning no relevant articles were missed by the tool. The mean specificity was 0.6095, indicating that MechaScreener correctly excluded 61.0% of irrelevant articles, significantly narrowing the pool for human review.

An individual breakdown of each evaluation library is presented in Table 4.

Table 4: Results Across 10 Evaluation Libraries

Library Name	No. of Refs	Include Studies Correctly Included	Exclude Studies Correctly Excluded
Balneotherapy for Chronic Venous	327	9/9 (100%)	259/318 (81%)
Calcium Vitamin D for Bones	5,493	6/6 (100%)	1,825/5,487 (33%)
Methylxanthine for Apnea	2,446	28/28 (100%)	1,422/2,418 (59%)
Phosphodiesterase 5 Inhibitors	2,333	6/6 (100%)	2,128/2,327 (91%)
Topical and Oral Steroids for Om	3,706	28/28 (100%)	3,110/3,678 (85%)
Covid Schools	2,141	30/30 (100%)	439/2,111 (21%)
Diabetes Tb	5,011	48/48 (100%)	1,948/4,963 (39%)
Falls Prevention	17,000	9/9 (100%)	8,102/16,991 (48%)
Fluoride Fluorosis	9,244	40/40 (100%)	6,357/9,204 (69%)
Sanitation Diarrhoea	10,473	31/31 (100%)	8,684/10,442 (83%)

4 Discussion

In the context of systematic reviews (SRs), recall is universally recognised as the most critical metric; inappropriately excluding a single relevant study can introduce bias and potentially invalidate the findings of the entire review. Our results demonstrate that MechaScreener, when prompted with a structured 1–5 scoring system, can act as a highly effective and safe “first-pass” filter.

The primary value of MechaScreener lies in its ability to consistently maintain a perfect mean recall (1.0000) across all evaluated libraries while safely removing an average of approximately 61% of the irrelevant literature (mean specificity of 0.6095). This translates to a substantial reduction in the manual screening workload for researchers, reallocating valuable time and human resources towards downstream tasks such as full-text review, data extraction, and risk of bias assessment.

4.1 Performance Variance and Specificity

While the recall remained stable at 100% across both the development and evaluation datasets, we observed notable variance in specificity depending on the systematic review topic and study design. This variance demonstrates that the LLM’s capacity to confidently discard irrelevant literature is highly dependent on the precision of the research question and the standardisation of the field’s terminology.

For instance, highly specific pharmacological interventions such as *Phosphodiesterase 5 Inhibitors* [17] achieved a specificity of 0.9145, effectively eliminating over 90% of irrelevant studies. Clinical trials in these domains typically feature highly structured abstracts and unambiguous PICOT elements, allowing the LLM to easily detect off-target papers.

Conversely, broader or more complex topics, particularly those involving non-randomised controlled trials (non-RCTs) like *Covid Schools* [19] or *Diabetes Tb* [20], exhibited lower specificities (0.2080 and 0.3925, respectively). These public health and observational fields often involve diverse study designs, varied outcome measures, and less structured reporting. When abstract information is vague, lacks explicit PICOT details, or touches on related thematic concepts without matching precise criteria, MechaScreener conservatively assigns a higher probability of inclusion. This built-in conservatism prevents the accidental exclusion of relevant studies, correctly prioritising a perfect recall over maximal workload reduction when faced with ambiguity.

4.2 Limitations

Despite the robust performance, several limitations must be acknowledged. First, as noted in the methodology, the evaluation dataset’s ground truth for ‘relevant’ studies was based on final full-text inclusion decisions rather than initial human title and abstract screening decisions. Because abstracts often lack complete methodological details, an automated title and abstract screener may correctly flag a study as potentially relevant based on the abstract alone, only for it to be counted as a false positive because the study was ultimately excluded during the human full-text review phase. Consequently, the reported specificity may artificially underestimate the tool’s true precision at the abstract screening level.

Second, the current methodology relies heavily on the availability of study abstracts. As highlighted in Table 3, thousands of references across the tested libraries (e.g. over 2,500 in the *Fluoride Fluorosis* library) lacked abstract data. Studies without abstracts are inherently difficult to screen automatically and often default to human review, reducing the overall workload savings.

Third, MechaScreener is designed explicitly as a title and abstract screening assistant. It does not perform full-text screening, meaning human reviewers must still assess the remaining literature in full detail. Finally, while we utilised a deterministic approach (i.e. `temperature = 0`) to ensure reproducibility, the system’s performance remains fundamentally tied to the quality, clarity, and comprehensiveness of the user-provided eligibility criteria. Poorly defined criteria may lead to lower specificity as the model struggles to differentiate borderline studies.

4.3 Future Directions

Future research should explore the integration of MechaScreener directly into standard systematic review workflows and reference management platforms. Evaluating its performance across a broader range of foundational LLMs, as well as refining the prompt architecture to better handle title-only references, are key next steps. Furthermore, expanding the system’s capabilities to assist with automated full-text screening and data extraction could further streamline the evidence synthesis pipeline. Overall, these findings solidify the role of LLMs as powerful, dependable augmentations to traditional systematic review methodologies.

5 Conclusion

MechaScreener provides a reliable method for screening research articles. By leveraging large-scale LLMs and a probabilistic scoring approach, researchers can use MechaScreener to reduce their screening workload by more than half while maintaining the rigorous standards required for systematic literature synthesis.

Data and Code Availability

The dataset (relevance scores generated by Mechascreener and ground truth include/exclude decisions), source code for the analysis, along with the scripts used to generate the figures and tables presented in this paper, are available in the GitHub repository:

<https://github.com/IEBH/mechascreener-paper>

References

- [1] Justin Clark, Paul Glasziou, Chris Del Mar, Alexandra Bannach-Brown, Paulina Stehlik, and Anna Mae Scott. A full systematic review was completed in 2 weeks using automation tools: a case study. *Journal of Clinical Epidemiology*, 121:81–90, May 2020.

- [2] Rohit Borah, Andrew W. Brown, Patrice L. Capers, and Kathryn A. Kaiser. Analysis of the time and workers needed to conduct systematic reviews of medical interventions using data from the PROSPERO registry. *BMJ open*, 7(2):e012545, February 2017.
- [3] Matthew J. Page, Joanne E. McKenzie, Patrick M. Bossuyt, Isabelle Boutron, Tammy C. Hoffmann, Cynthia D. Mulrow, Larissa Shamseer, Jennifer M. Tetzlaff, Elie A. Akl, Sue E. Brennan, Roger Chou, Julie Glanville, Jeremy M. Grimshaw, Asbjørn Hróbjartsson, Manoj M. Lalu, Tianjing Li, Elizabeth W. Loder, Evan Mayo-Wilson, Steve McDonald, Luke A. McGuinness, Lesley A. Stewart, James Thomas, Andrea C. Tricco, Vivian A. Welch, Penny Whiting, and David Moher. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ (Clinical research ed.)*, 372:n71, March 2021.
- [4] Wichor M. Bramer, Jelena Milic, and Frans Mast. Reviewing retrieved references for inclusion in systematic reviews using EndNote. *Journal of the Medical Library Association: JMLA*, 105(1):84–87, January 2017.
- [5] Fernando M. Delgado-Chaves, Matthew J. Jennings, Antonio Atalaia, Justus Wolff, Rita Horvath, Zeinab M. Mamdouh, Jan Baumbach, and Linda Baumbach. Transforming literature screening: The emerging role of large language models in systematic reviews. *Proceedings of the National Academy of Sciences*, 122(2):e2411962122, January 2025.
- [6] Justin Clark, Belinda Barton, Loai Albarqouni, Oyungerel Byambasuren, Tanisha Jowsey, Justin Keogh, Tian Liang, Christian Moro, Hayley O’Neill, and Mark Jones. Generative artificial intelligence use in evidence synthesis: A systematic review. *Research Synthesis Methods*, 16(4):601–619, July 2025.
- [7] Institute for Evidence Based Healthcare. TERA Mechascreeener. <https://tera-tools.com/mechascreeener>. Accessed: 2026-03-19.
- [8] Rebecca A. Speckman and Janna L. Friedly. Asking Structured, Answerable Clinical Questions Using the Population, Intervention/Comparator, Outcome (PICO) Framework. *PM&R*, 11(5):548–553, May 2019.
- [9] Mina Bakhit, Emma Baillie, Natalia Krzyzaniak, Mieke van Driel, Justin Clark, Paul Glasziou, and Christopher Del Mar. Antibiotic prescribing for acute infections in synchronous telehealth consultations: a systematic review and meta-analysis. *BJGP open*, 5(6):BJGPO.2021.0106, 2021.
- [10] Oyungerel Byambasuren, Paulina Stehlik, Justin Clark, Kylie Alcorn, and Paul Glasziou. Effect of covid-19 vaccination on long covid: systematic review. *BMJ medicine*, 2(1):e000385, 2023.
- [11] Kwame Peprah Boaitey, Mina Bakhit, and Tammy C. Hoffmann. Mapping the evidence about the natural history of acute infections commonly seen in primary care and managed with antibiotics: a scoping review. *BMC infectious diseases*, 24(1):721, July 2024.
- [12] Hannah Greenwood, Alexandra R. Davidson, Rae Thomas, and Loai Albarqouni. Common barriers and enablers to the use of non-drug interventions for managing common chronic conditions in primary care: an overview of reviews. *BMC primary care*, 25(1):108, April 2024.
- [13] Hannah Greenwood, Katelyn Barnes, Justin Clark, Lauren Ball, and Loai Albarqouni. Long-Term Effect of Salt Substitution for Cardiovascular Outcomes : A Systematic Review and Meta-analysis. *Annals of Internal Medicine*, 177(5):643–655, May 2024.

- [14] Melissa Andreia de Moraes Silva, Luis Cu Nakano, Lgia L. Cisneros, and Fausto Miranda. Balneotherapy for chronic venous insufficiency. *The Cochrane Database of Systematic Reviews*, 1(1):CD013085, January 2023.
- [15] Luca Mndez-Snchez, Patricia Clark, Tania M. Winzenberg, Peter Tugwell, Paulina Correa-Burrows, and Rebecca Costello. Calcium and vitamin D for increasing bone mineral density in premenopausal women. *The Cochrane Database of Systematic Reviews*, 1(1):CD012664, January 2023.
- [16] Keri A. Marques, Matteo Bruschetti, Charles C. Roehr, Peter G. Davis, Michelle Fiander, and Roger Soll. Methylxanthine for the prevention and treatment of apnea in preterm infants. *The Cochrane Database of Systematic Reviews*, 10(10):CD013830, October 2023.
- [17] Nancy Maltez, Lara J. Maxwell, Fadumo Rirash, Elizabeth Tanjong Ghogomu, Sarah E. Harding, Paul C. Tingey, George A. Wells, Peter Tugwell, and Janet Pope. Phosphodiesterase 5 inhibitors (PDE5i) for the treatment of Raynaud’s phenomenon. *The Cochrane Database of Systematic Reviews*, 11(11):CD014089, November 2023.
- [18] Caroline A. Mulvaney, Kevin Galbraith, Katie E. Webster, Mridul Rana, Rachel Connolly, Ben Tudor-Green, Tal Marom, Mat Daniel, Roderick P. Venekamp, Anne Gm Schilder, and Samuel MacKeith. Topical and oral steroids for otitis media with effusion (OME) in children. *The Cochrane Database of Systematic Reviews*, 12(12):CD015255, December 2023.
- [19] Hannah Littlecott, Shari Krishnaratne, Jacob Burns, Eva Rehfuss, Kerstin Sell, Carmen Klinger, Brigitte Strahwald, Ani Movsisyan, Maria-Inti Metzendorf, Petra Schoenweger, Stephan Voss, Michaela Coenen, Roxana Mller-Eberstein, and Lisa M. Pfadenhauer. Measures implemented in the school setting to contain the COVID-19 pandemic. *The Cochrane Database of Systematic Reviews*, 5(5):CD015029, May 2024.
- [20] Juan Va Franco, Brenda Bongaerts, Maria-Inti Metzendorf, Agostina Risso, Yang Guo, Laura Pea Silva, Melanie Boeckmann, Sabrina Schlesinger, Johanna Aag Damen, Bernd Richter, Annabel Baddeley, Mathieu Bastard, Anna Carlqvist, Maria Nieves Garcia-Casal, Bianca Hemmingsen, Farai Mavhunga, Jennifer Manne-Goehler, and Kerri Viney. Diabetes as a risk factor for tuberculosis disease. *The Cochrane Database of Systematic Reviews*, 8(8):CD016013, August 2024.
- [21] Sharon R. Lewis, Lisa McGarrigle, Michael W. Pritchard, Alessandro Bosco, Yang Yang, Ashley Gluchowski, Jana Sremanakova, Elisabeth R. Boulton, Matthew Gittins, Anneliese Spinks, Kilian Rapp, Daniel E. MacIntyre, Roderick J. McClure, and Chris Todd. Population-based interventions for preventing falls and fall-related injuries in older people. *The Cochrane Database of Systematic Reviews*, 1(1):CD013789, January 2024.
- [22] May Chun Mei Wong, Rui Zhang, Bella Weijia Luo, Anne-Marie Glenney, Helen V. Worthington, and Edward Chin Man Lo. Topical fluoride as a cause of dental fluorosis in children. *The Cochrane Database of Systematic Reviews*, 6(6):CD007693, June 2024.
- [23] Valerie Bauza, Wenlu Ye, Jiawen Liao, Fiona Majorin, and Thomas Clasen. Interventions to improve sanitation for preventing diarrhoea. *The Cochrane Database of Systematic Reviews*, 1(1):CD013328, January 2023.